

Theory 2. Convex sets and functions.

All tasks cost 10/8 points (so the total sum for all tasks excluding bonus is 10).

Task 1 Suppose that C is a convex set in some linear space. Show that (a) closure $\overline{C}$ and (b) interior $\text{int}(C)$ of the set C are also convex sets.

Task 2 Let C and D be some sets in linear space. Show that:

- (a) $\text{Conv}(C \cup D) = \text{Conv}(\text{Conv}(C) \cup \text{Conv}(D))$.
- (b) $\text{Conv}(C \cap D) \subseteq \text{Conv}(C) \cap \text{Conv}(D)$.

Task 3 Show that the following set is convex: $\{x \in \mathbb{R}^n : x^T P x \leq (c^T x)^2, c^T x \geq 0\}$, where $P \in S_{++}^n$, $c \in \mathbb{R}^n$.

Task 4 Show that $\text{Conv}\{xx^T : x \in \mathbb{R}^n, \|x\|_2 = 1\} = \{A \in S_+^n : \text{tr}(A) = 1\}$.

Task 5 Prove the differential criterion of strong convexity: a differentiable function f is μ strongly convex if and only if $f(y) \geq f(x) + \nabla f(x)^T(y - x) + \frac{\mu}{2}\|y - x\|^2$ for all $x, y \in \text{dom} f$. The function f is called μ strongly convex if the function $f(x) - \frac{\mu}{2}\|x\|^2$ is convex.

Task 6 Using standard examples of convex functions and operations preserving convexity, show that the following functions are convex:

- (a) $f : \mathbb{R}^n \rightarrow \mathbb{R}$ – function $f(x) = \max_{1 \leq i \leq n} c_i \log(1 + \exp(|x_i|))$, where $c_1, \dots, c_n \geq 0$.
- (b) $f : \mathbb{R}^n \rightarrow \mathbb{R}$ – function $f(x) = \log(\sum_{i=1}^n \exp(\max^2\{0, x_i\}))$.

Task 7 Let's consider the function $f : \mathbb{R}_{++}^n \rightarrow \mathbb{R}$, $f(x) = \prod_{i=1}^n x_i^{\alpha_i}$, $\alpha_1, \dots, \alpha_n \geq 0$, $\sum_{i=1}^n \alpha_i = 1$. Show that this function is concave.

Task 8 Show that the following function is convex:

$$f(x) = \frac{1}{x_1 - \frac{1}{x_2 - \frac{1}{\ddots - \frac{1}{x_n}}}}.$$

Here the function is defined on a subset of $\mathbb{R}^n$, where every denominator is strictly positive.

Bonus task (2 pts) Prove that continuous function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if and only if

$$f\left(\frac{x+y}{2}\right) \leq \frac{f(x) + f(y)}{2}, \quad \forall x, y \in \mathbb{R}^n.$$